



Breakthrough achieved in submarine cable transmission

Using a new transponder, NEC Corporation has effectively concluded a field test for an optical submarine cable system over a long distance. Based on NEC's research, this transponder achieves



Field trial route (Credit: www.nec.com)

a global-leading transmission performance of 800 gigabits per second (Gbps). In a field trial using the Indonesia Global Gateway (IGG) optical submarine cable owned by PT Telkom Indonesia, the country's premier telecommunications provider, NEC deployed its latest XF3200 transponder. This test resulted in an unprecedented optical transmission of 800Gbps signals over 2,100km. The XF3200 outperformed, delivering a transmission capacity about 30% higher than NEC's current products, meeting the rising demands of global communication. The transponder's advanced design ensures space-efficiency, energy conservation, high scalability, and adaptable maintenance, collectively reducing total ownership costs.

Prototype chips deliver AI task solutions with minimal power consumption

IBM has introduced an advancement in artificial intelligence (AI) with the creation of energy-efficient analogue AI chips. Demonstrating an estimated 14 times higher energy efficiency than traditional digital AI chips, this innovation holds significant potential in reducing the carbon footprint of AI operations. By leveraging the power efficiency of analogue chips, IBM's prototype is particularly tailored for speech recognition and transcription tasks. In tests, these analogue chips delivered comparable performance, speed, and efficiency as digital alternatives. Unlike earlier versions, IBM's design can accommodate expansive AI models, supporting up to 17 million parameters. These chips streamline operations by executing multiply-accumulate (MAC) functions directly within memory, optimising energy use. In evaluations, the chips excelled in keyword detection and transcription tasks. Beyond language processing, the chips also showcase capabilities in computer vision and image recognition.

Robot wheelchair technology assists people with lower-limb disabilities

The Korea Institute of Machinery and Materials (KIMM) has unveiled a robot wheelchair with innovative features to assist people with lower-limb disabilities. The team has developed a stair-climbing module for wheelchairs, featuring a concealed '≡'-shaped crawler, akin to tank tracks, which



A robot wheelchair descending stairs (Credit: www.businesskorea.co.kr)

activates for stair navigation, eliminating the slipping issue common in conventional designs. Additionally, the standing module incorporates a unique parallelogram structure and self-weight compensation technology, reducing the required motor power by 80%. This allows users to adopt distinct postures—

standing, lying back, leaning, and height adjustment—to enhance comfort and circulation, potentially warding off bedsores. These modules can operate independently or as one unit, enabling stair-climbing, standing, and posture adjustments, greatly improving mobility for those with lower-limb disabilities.

Self-healing wearables powered by liquid metal circuitry

National University of Singapore (NUS) scientists have recently developed an ultra-flexible, self-healing, and highly conductive substance ideal for stretchable electronic circuits. Using the innovative Bilayer Liquid-Solid Conductor (BiLiSC) liquid metal circuitry, electronic devices can withstand significant deformations, self-repair, and maintain consistent performance, making it an ideal material for wearable technologies that conform to the body's movements. The team has developed a cost-effective, scalable production method for BiLiSC and has demonstrated its versatility in creating essential wearable electronic components like pressure sensors, interconnects, and antennas. In tests, a BiLiSC-integrated robotic arm detected subtle pressure changes efficiently and maintained signal integrity during complex movements.